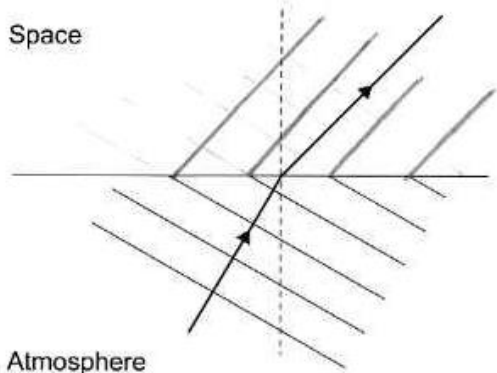


				 No marks							
	(c)	(i)		Satellite orbits in the same time / 24 hours (1) that it takes the earth to spin once (1) Accept for 1 mark they both orbit in 24 hours	2			2			
		(ii)		Total distance = $4 \times (1)$ [$3.6 \times 10^7 = 14.4 \times 10^7$ [m]] Substitution: $3 \times 10^8 = \frac{14.4 \times 10^7 \text{ecf}}{\text{time}}$ (1) Time = 0.48 [s] (1) Award 2 marks for answers of 48×10^n or 0.24 [s] Award 1 mark for answers of 24×10^n or 0.12 [s]	1	1		3	3		
		(iii)		Distance travelled would be the same or both have 4 steps or still use 2 satellites [so disagree]			1	1			
				Question 2 total	3	9	1	13	7	0	